

ZigBee

A wireless communication standard able to connect smart devices. It could be used with Industrial control and monitoring, etc.

The main features of this device are given by:

- Reliability
- Low costs: ZigBee modules are simple and light, not consuming a lot of energy.
- Long battery life, also due to limited power consumption, with nodes being almost all the time in sleep state, and they activate when they need to transmit data.
- Low complexity network
- Use of free frequencies: unlicensed frequencies are used, and so no payment is required. It uses 2.4 GHz.
- Support of high number of nodes: up to 65k nodes.
- Interoperability: devices can communicate with each other
- Security: authentication phases, key management.

The protocol stack is given by a lighter version with respect to other standards, and for PHY and MAC layers we use 802.15.4, while for Application layer a specific ZigBee application is used.

ZigBee relies on the direct sequence spread spectrum (DSSS) modulation. The idea is to create a signal that occupies a large bandwidth, which would imply large capacity and higher data rates. In order to have these devices access the channel the CSMA/CA protocol is used, and the different devices can choose a particular time slot to avoid collisions of packages.

The coverage range depend on different parameters, and it can reach up to 1000 meters, considering that LoS is guaranteed.

It also allows us to have mesh topology, with devices being able to communicate with each other, but the transmission has to be optimised, with one like AODV-type routing techniques. This allows low data rate.

ZigBee uses different topologies:

- Star topology: there is a central coordinator node, with everyone connected to it
- Peer-to-Peer topology: each device can communicate with other nodes. This can evolve into a mesh and cluster-tree topology. In particular, for the cluster-

tree there is just one path between two nodes, and nodes have to communicate with the PAN coordinator.

AODV, in particular, is only used on demand.

The roles the node can have are given by:

- The FDD (full function device) has to be connected with the PAN coordinator, and can also be connected to a Reduced function device.
- The Reduced function device (RFD) is connected to either the PAN coordinator or the FDD, depending on the topology.
- The PAN coordinator.

In order to create a network, specifically for a cluster-tree topology, we have a node asking other devices to access the network itself through a request. If it's accepted, this new node will be connected to the FDD. The distance between the child and the PAN coordinator will be one hop. This creates a hierarchical network structure.

For the star topology we have the full function device having to send beacon requests to find some nodes which could be considered the PAN coordinator. If the PAN coordinator is present in the network it can respond with a response beacon. In order to support a high number of nodes in the network, addresses are only 16 bits, and these addresses are sent to nodes from the PAN coordinator, which has a table of addresses sent to FDDs, and these "parents" are then going to transmit addresses to sub nodes of the topology. For this reasons, we need to consider the routing mechanism.

It could be hierarchical, so that messages are sent from the PAN coordinator to FDD, and then to RDD, or through a "best-path" mechanism.

The routing technique used specifically as we mentioned is the AODV. It's activated just once, when the node wan to send (??) data? In addition to this, we also have other protocols used to determine the best possible path. The main things done by AODV are the route discovery, so determining the path used to link a source node to a destination node, and also route maintenance, as if an error occurs in a link between the nodes, the connection is cancelled.

Let's consider a node that wants to transmit data to a destination node, but does not know how. For this, it sends ...

The destination node, or other nodes that know how to reach the destination, will send back a response telling the device the different paths the source could use. Each request has a TTL, as well as a back-off mechanism.

The bands for ZigBee are given by the one used for Wi-Fi, so the 2.4 GHz, but also other bands given by 915-921 MHz, and also 868 MHz, with small data rates and also small number of channels (16 for the bandwidth of the Wi-Fi, 10 for USA bandwidth, and 1 for EU).

The distances vary depending on channel conditions, like LoS.

The TX power is fixed.

When ZigBee uses the bandwidth of the Wi-Fi interferences can occur. Channels are divided and non-overlapping channels are given by 1,6,11.

If we compare with Bluetooth technology, we can still have interferences, but the probability is actually really small, given by $\frac{3}{79}$.

🔗 Exercise 1

Exercise



In a ZigBee mesh network for an industrial automation system, the following specifications are provided:

Parameter	Value
Total number of nodes	45 nodes (1 coordinator + 12 FFD routers + 32 RFD end devices)
Data rate	250 kbps (2.4 GHz band)
Data packet size	80 bytes
ACK packet size	15 bytes
Propagation time per hop	0.5 ms
Processing time per node	2 ms
Maximum allowed hops	8 hops
Network size	200 m x 150 m
Average distance between adjacent nodes	25 m

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Exercise

1. Calculate the total latency for a packet crossing 5 hops.

- Propagation time per hop: 0.5 ms
- Processing time per node: 2 ms
- How many nodes process the packet in 5 hops?
- How much time does the packet take from source to destination?

2. Calculate the total transmission time for a packet with ACK.

- Transmit an 80-byte packet at 250 kbps
- Receive a 15-byte ACK at 250 kbps
- SIF (Short Interframe Space) = 192 μ s
- What is the total time for a complete transmission (data + ACK)?

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The number of nodes that process the packet are 6, as we have 5 hops plus the destination node.

The total propagation time is given by $6 \text{ hops} \times 0.5 \frac{\text{ms}}{\text{hop}} = 2.5 \text{ ms}$.

The total processing time is $6 \text{ nodes} \times 2 \text{ ms/node} = 12 \text{ ms}$.

The total latency is $2.5 \text{ ms} + 12 \text{ ms} = 14.5 \text{ ms}$.

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The packet duration is derived by multiplying the 80 bytes and the 8 bits per byte, so 640 bits. The Time is 640 bits over 250 k bits per second, and so 2.56 ms.

The ACK packet duration is 15 bytes multiplied by 8 bits per byte, and so 120 bits. The time is the quantity over 250 k bits per second, and so 0.48 ms.

The SIF is 192 μ s and so 0.192 ms. Finally, the total time is $2.56 \text{ ms} + 0.192 \text{ ms} + 0.48 \text{ ms}$ so 3.232 ms.

🔍 Exercise 2

How many packets can be transmitted per second on a single channel?

If 10 nodes transmit simultaneously, what is the effective throughput per node?

Calculate the maximum network throughput.

The packets per second are given by 1 over 0.003232 seconds, so 309 packets per second.

The theoretical throughput is 309 packets multiplied by 640 bits per packet so approximately 198 kbps.

The efficiency is 198 kbps divided by 250 kbps, so 79.2%.

With 10 nodes transmitting simultaneously, we have the throughput being 198 kbps divided by 10, so 19.8 kbps.

🔗 Exercise 3

Calculate energy consumption per transmission:

- Transmission power: 100 mW
- Transmission duration: 3.232 ms
- Energy transmission in Joules: $E = P \times t$.
- If a node transmits 50 times a day, how many Joules/day does it consume?

The power is 100 mW so 0.1 W. The energy transmission is $0.1 \text{ W} \times 0.003232$ so 0.3232 mJ. The daily energy is thus this value multiplied by 50.

🔗 Exercise 4

Verify if the network is congested:

- 32 sensors send data every 30 seconds.
- Transmission duration per packet: 0.003232 seconds.
- How much time is the channel occupied in one minute?
- Is the channel saturated? ($> 100\%$ utilization)

64 packets are sent every minute. The time occupied for minute is 64 multiplied for transmission time, so 0.2068 seconds

The channel is utilized just at 0.2068 divided by 60 seconds, so 0.345 %.